



Donatien Schmitz

Software Engineer

Contact Details

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- in LinkedIn
- o Belgium / Remote

Education

PhD CS & Engineering

UCLouvain, Belgium

2021 – 2026

Resource Mgmt in Distributed
Stream Processing

Master Computer Science

UCLouvain, Belgium

2019 – 2021

Security & Networks

Bachelor Computer Science

UCLouvain, Belgium

2014 – 2019

Minor: Entrepreneurship

Skills

Languages

Python, Java, PHP,
JavaScript, Bash, SQL

Distributed Systems

Apache Flink,
Elastic Scaling, Resource Mgmt

Infrastructure

Docker, Kubernetes, Helm,
CI/CD, Envoy Proxy, Linux

Observability

Jupyter/Papermill,
Prometheus, Grafana

Backend / Web

Laravel/PHP, REST APIs,
MySQL, PostgreSQL, OpenMRS

Languages

French Native

English C1

Summary

Software engineer with 5+ years building distributed systems, data pipelines, and production infrastructure. PhD (UCLouvain, 2026) focused on resource management for Apache Flink and Kafka. Co-founded Read&Rate (2017–2021) as CTO — led all engineering from architecture to deployment. Open to backend / platform / DevOps roles.

Work Experience

Software Engineer — Post-Doctoral Researcher

Mar 2026 – Present

UCLouvain · LIME Lab (CS Applied to Medicine)

- Engineering software at the intersection of CS and healthcare.
- Contributing to OpenMRS clinical platform; integrating FHIR/HL7.
- Containerised deployments and CI/CD pipelines for research infra.

PhD Researcher — Distributed Systems Engineer

2021 – 2026

UCLouvain · ICTTEAM

- Built **Justin** [IFIP DAIS'25 Best Paper] [IFIP DisCoTec'25 Best Paper], **Charon**, **Alloy**, **StreamBed**: production-grade resource management systems on Flink.
- Large-scale experiments on Grid5000 & Kubernetes; reproducible benchmarks with Jupyter/Papermill.
- Teaching Assistant: Cloud Computing & Multicore Programming (2021–2025).
- Industry collaboration with Digazu: real-world data lake workload analysis.

CTO & Co-founder

2017 – 2021

Read&Rate

- Built the entire stack for a community-driven publishing startup from scratch.
- Owned architecture, hiring, product decisions, and all infrastructure.
- Delivered a full web platform (Laravel/PHP + MySQL) with an active user community.

Treasurer

2026

Revue des Ingénieurs

- Managed the financial operations of a cultural student project with €50,000 annual cashflow.

Selected Projects

Justin — Elastic Flink Scaler [IFIP DAIS'25 Best Paper] [IFIP DisCoTec'25 Best Paper]

Hybrid CPU/memory auto-scaler for Flink — resizes task slots according to operator needs. [github](#)

Charon — Multi-Query Scheduler

Fine-grain resource allocator and task placement optimizer for multi-query Flink. [github](#)

Alloy — Transparent Storage Proxy

Envoy-based proxy decoupling Flink pipelines from storage backends. [github](#)

StreamBed — Capacity Planner

Predicts stream processing cluster resource requirements from telemetry. [github](#)

Read&Rate — Publishing Platform

Community-driven platform built from scratch as CTO co-founder (2017–2021).